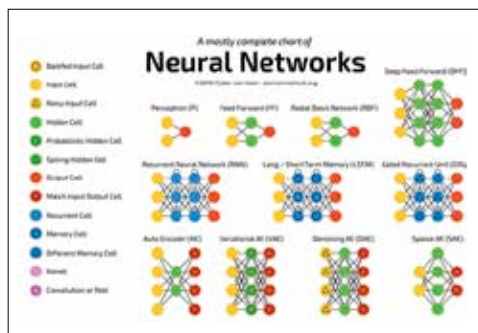


Hidden Layers: Between the input and output nodes, analysts have to specify the number of hidden layers that populate the neural network. A general rule of thumb would say that adding more layers is bound to make the program more accurate and successful, but this tends to add more pressure on the system to compute and confirm the values.

A major determinant in the accuracies is the learning rule, the number of epochs and learning rate which will be discussed later.

Types of neural networks

The neural network employed for any data problem depends on what the output-segmentation or classification, is desired to be and the format for the input. Convolutional neural networks, for example, would be more useful for processing and identifying images while a sigmoidal segmentation network would be more suited for data in industries.



Types of Neural Network Topologies

Feed Forward

ANN: A typical neural network program is one where the input leads straight to the output with any number of hidden layers. The main benefit of such neural network schemes is that they help recognise input patterns.

Feedback ANN:

As the name suggests, the output goes back into the network to learn how to obtain the best results. A feedback loop is inserted from the output to the input which acts as a new input.

This is well designed to solve engineering and optimisation problems, commonly those that range in large equation orders and complex components (like logs and exponents). They're best used for correcting errors in systems.

Modular Neural Networks: These happen to be a collection of multiple networks functioning independently while connecting to the output. Each neural network has inputs that are specific to the node such that no nodes interact with each other.